

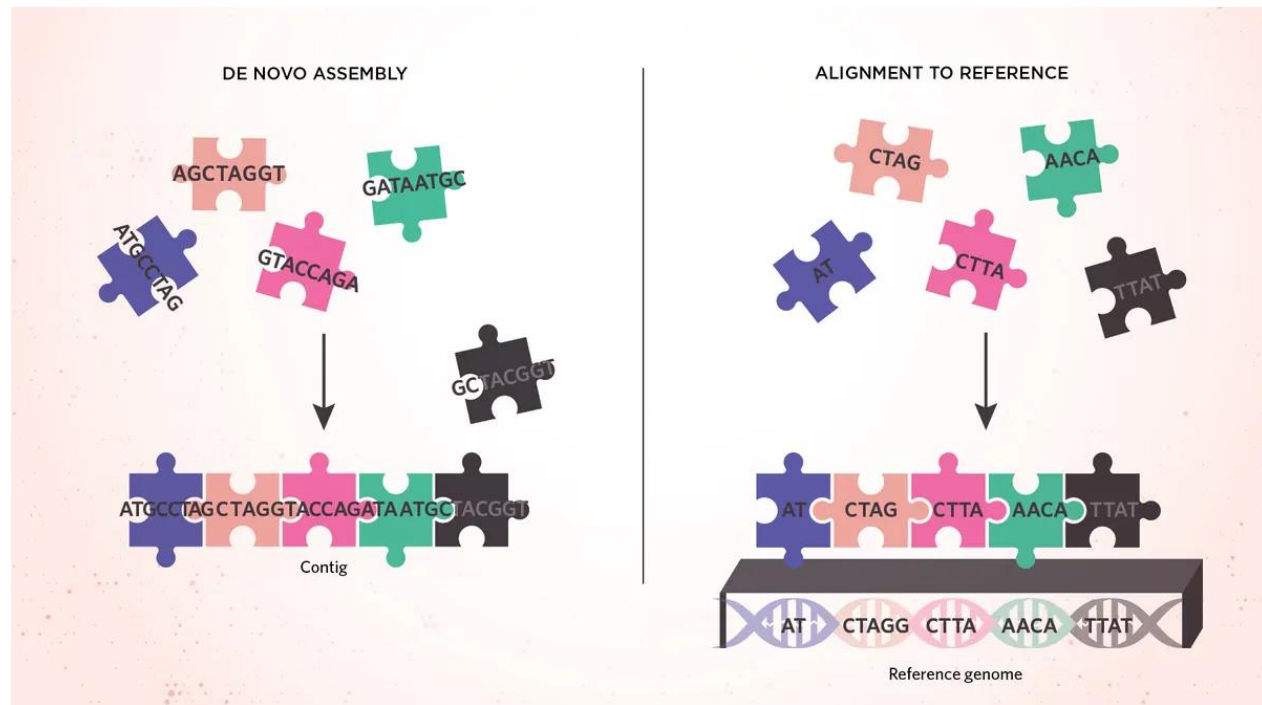
Learning Your Data: Data Processing of Genetic Data.

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What is Genome Build?

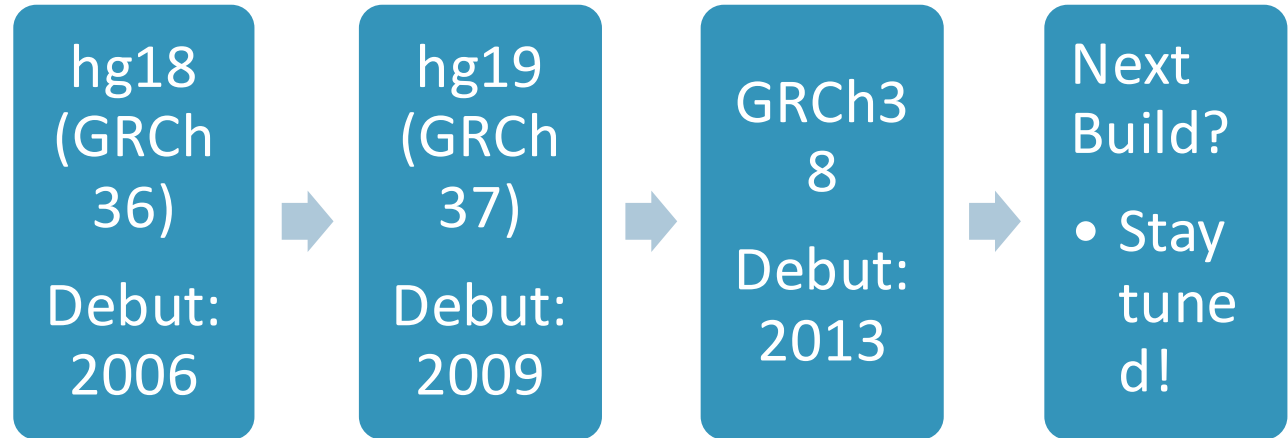
A human genome reference build is essentially a "standard map" of human DNA - it's the agreed-upon sequence of all 3 billion DNA letters that serves as the universal reference point for comparing everyone else's genomes



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What is a Genome Build?

As sequencing technology improves and we discover more about human genetic variation, the Human Reference Genome is updated to fix errors and fill in gaps. Each iteration of the Reference Human Genome is called a build!



- The builds essentially represent snapshots of our best knowledge at the time.
- As technology gets better, we can sequence harder regions, fix mistakes, and create more accurate reference maps for everyone to use.

Variant Calling

```
##fileformat=VCFv4.2
##fileDate=20250712
##source=PLINKv1.90
##contig=<ID=1,length=249218993>
##contig=<ID=2,length=243048761>
##contig=<ID=3,length=197833759>
##contig=<ID=4,length=190939666>
##contig=<ID=5,length=180696890>
##contig=<ID=6,length=170919471>
##contig=<ID=7,length=159119221>
##contig=<ID=8,length=146296415>
##contig=<ID=9,length=141066492>
##contig=<ID=10,length=135522247>
##contig=<ID=11,length=14940417>
```

Mandatory
Columns for
VCFs to be
correctly read

ID: The chromosome/contig

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	BenK_ID001
1	734462	rs12564807	G	A	.	.	PR	GT	0/0
1	752721	rs3131972	A	C	.	.	PR	GT	0/0
1	760998	rs148828841	C	A	.	.	PR	GT	0/0
1	776546	rs12124819	A	G	.	.	PR	GT	0/0
1	787173	rs115093905	G	A	.	.	PR	GT	./.

chromosomal positions and
associated metadata, making it
the go-to format for sharing
genomic variation data

```
##contig=<ID=24,length=5903210>
##contig=<ID=26,length=16538>
##INFO=<ID=PR,Number=0,Type=Flag,Description="Provisional reference allele, may not be based on real reference genome">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT BenK_ID001
1 734462 rs12564807 G A . . PR GT 0/0
1 752721 rs3131972 A C . . PR GT 0/0
1 760998 rs148828841 C A . . PR GT 0/0
1 776546 rs12124819 A G . . PR GT 0/0
1 787173 rs115093905 G A . . PR GT ./.
```

How do I know what Build my dataset is in?

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	BenK_ID001
1	734462	rs12564807	G	A	.	.	PR	GT	0/0
1	752721	rs3131972	A	C	.	.	PR	GT	0/0
1	760998	rs148828841	C	A	.	.	PR	GT	0/0
1	776546	rs12124819	A	G	.	.	PR	GT	0/0
1	787173	rs115093905	G	A	.	.	PR	GT	./.
1	798959	rs11240777	G	.	.	.	PR	GT	0/0

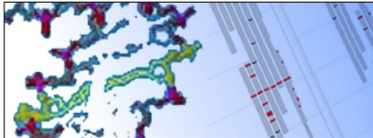
How do I know what Build my dataset is in?

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information [Log in](#)

Service Alert: Planned Maintenance beginning July 25th
Most services will be unavailable for 24+ hours starting 9 PM EDT. [Learn more about the maintenance.](#)

dbSNP [Advanced](#) [Search](#) [Help](#)

 **dbSNP**
dbSNP contains human single nucleotide variations, microsatellites, and small-scale insertions and deletions along with publication, population frequency, molecular consequence, and genomic and RefSeq mapping information for both common variations and clinical mutations.

Getting Started
[dbSNP 25th Anniversary](#)
[Overview of dbSNP](#)
[About Reference SNP \(rs\)](#)
[Factsheet](#)
[FAQ](#)
[Entrez Updates \(May 26, 2020\)](#)

Submission
[How to Submit](#)
[Hold Until Published \(HUP\) Policies](#)
[Submission Search](#)

Access Data
[Web Search](#)
[eUtils API](#)
[Variation Services](#)
[FTP Download](#)
[Tutorials on GitHub](#)

Important: When using dbSNP, please cite the resource using the following publication: [The evolution of dbSNP: 25 years of impact in genomic research.](#)

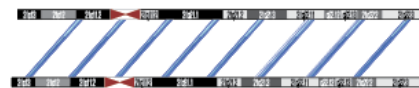
ALFA Project Release 4 with over 900M variants from 400K subjects is now available (May 15, 2025)
The goal is to provide allele frequency from more than 1 million dbGaP subjects with regular updates. Visit the project [page](#) for more information or view the introduction video below.

dbSNP

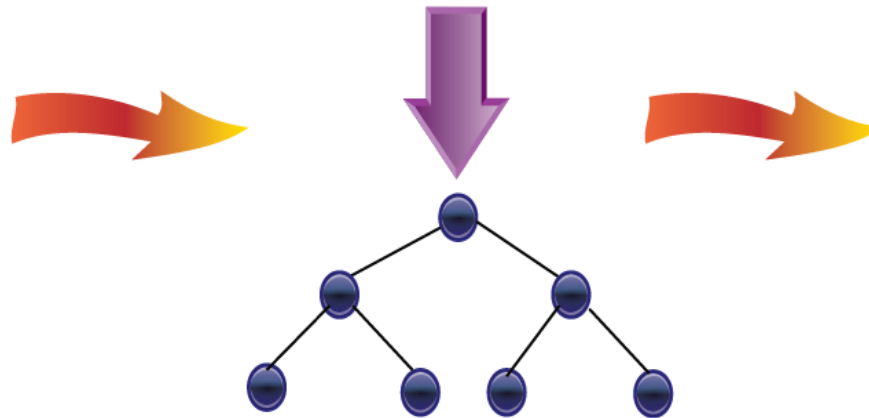
What is leftover?



Coordinate file based on
genome build version_1



UCSC .chain file



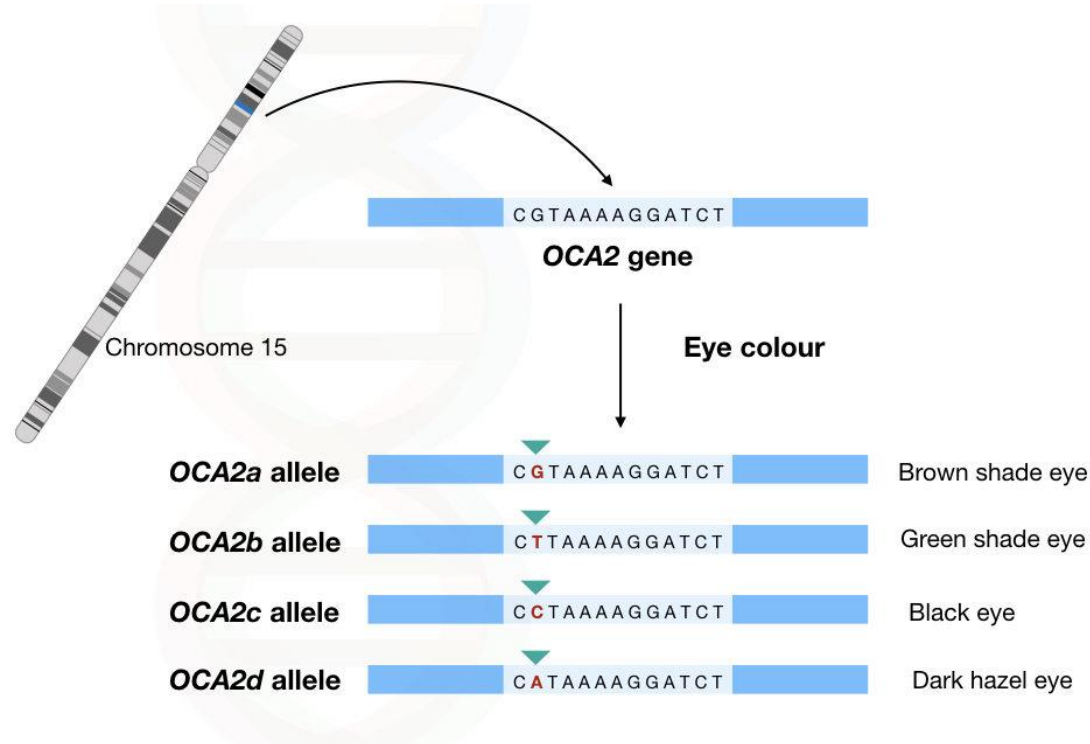
Interval Tree



Coordinate file based on
genome build version_2

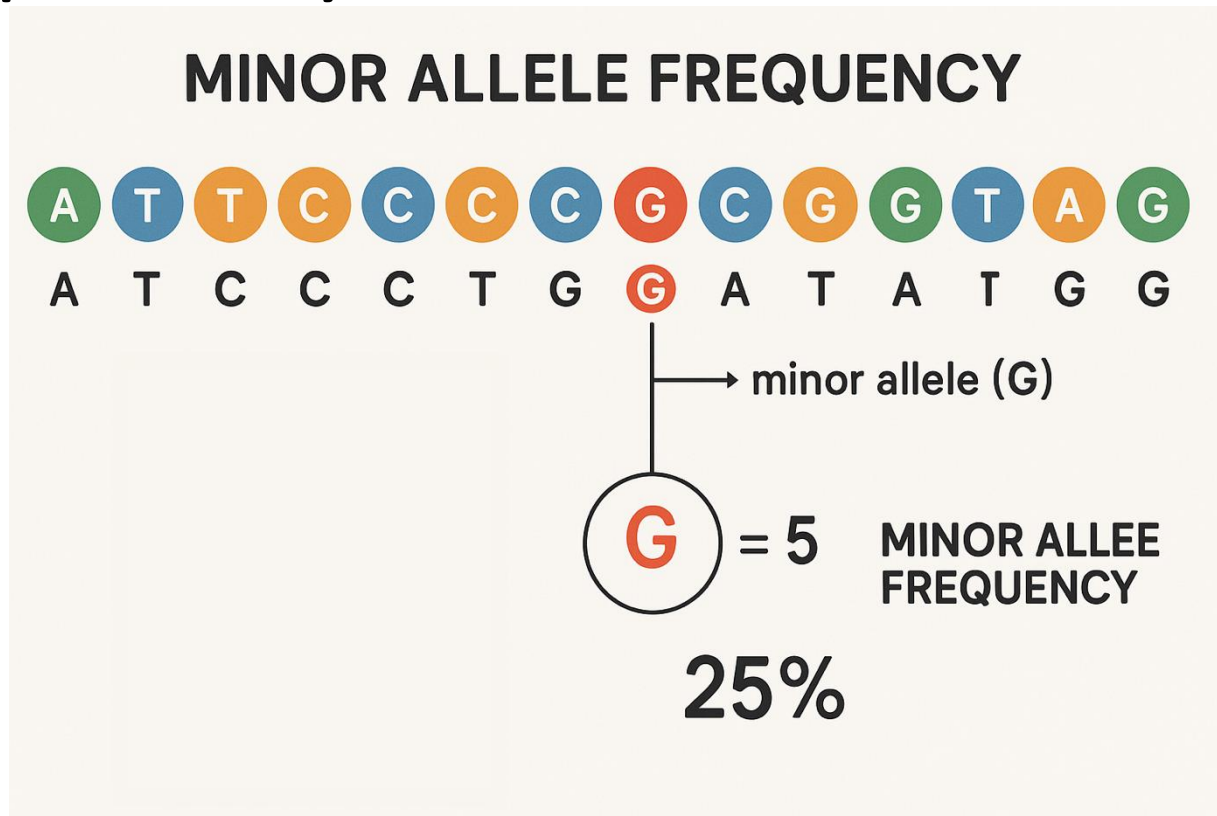
CrossMap

What do we mean by reference allele?



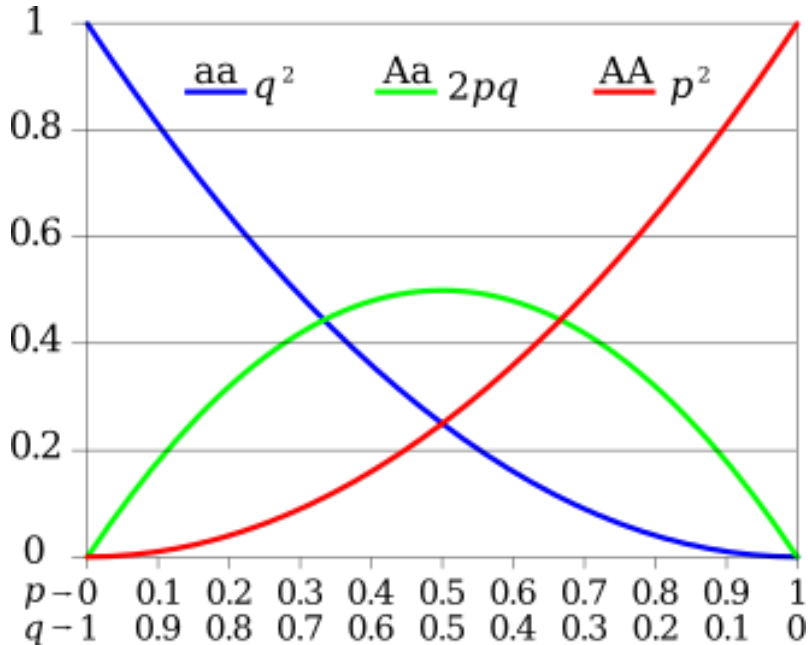
<https://geneticeducation.co.in/gene-vs-allele/>

What is Minor Allele Frequency?



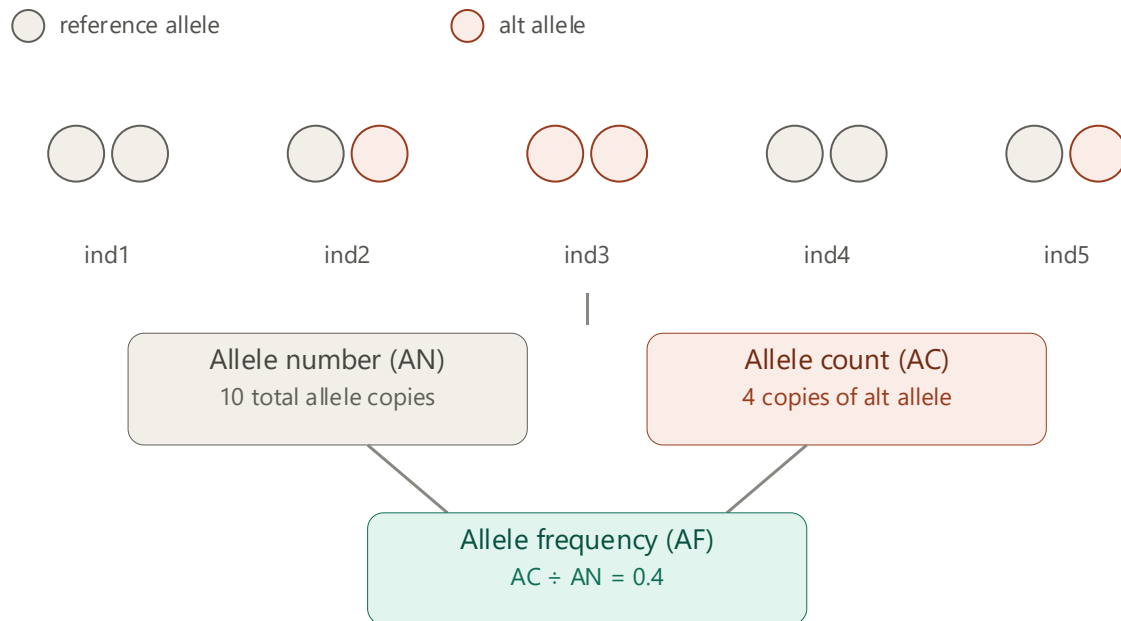
Hardy-Weinberg Equilibrium

- Hardy-Weinberg equilibrium describes the relationship between allele frequencies and genotype frequencies in an idealized population
 - Does it really exist?
 - Not a good approximation statistically
 - No gene flow
 - Infinite population size
 - Random mating
 - No natural selection



Allele Number, Allele Count, and Allele frequency

What's the difference?



PLINK!

(Pedigree-linkage)

Input set up PLINK

```
> Plink --bfile/vcf INSERT
```

bfile: PLINK formatting 3
separate files

.fam

.bed

.bim

```
--real-ref-alleles
```

```
--keep-allele-order
```

Common QC commands in PLINK

--geno: the amount of missing per variant

--mind: the amount of missing per sample

--hwe: p-value of how much a variant
violates HWE

--maf: Remove variants with a MAF below
your given threshold

--keep: Only retain a list of samples

--remove: Remove a list of samples

--extract: Only keep a list of variants or
genomic coordinates

--exclude: Remove a list of variants or
genomic coordinates

output set up PLINK

```
--make-bed
```

```
--recode
```

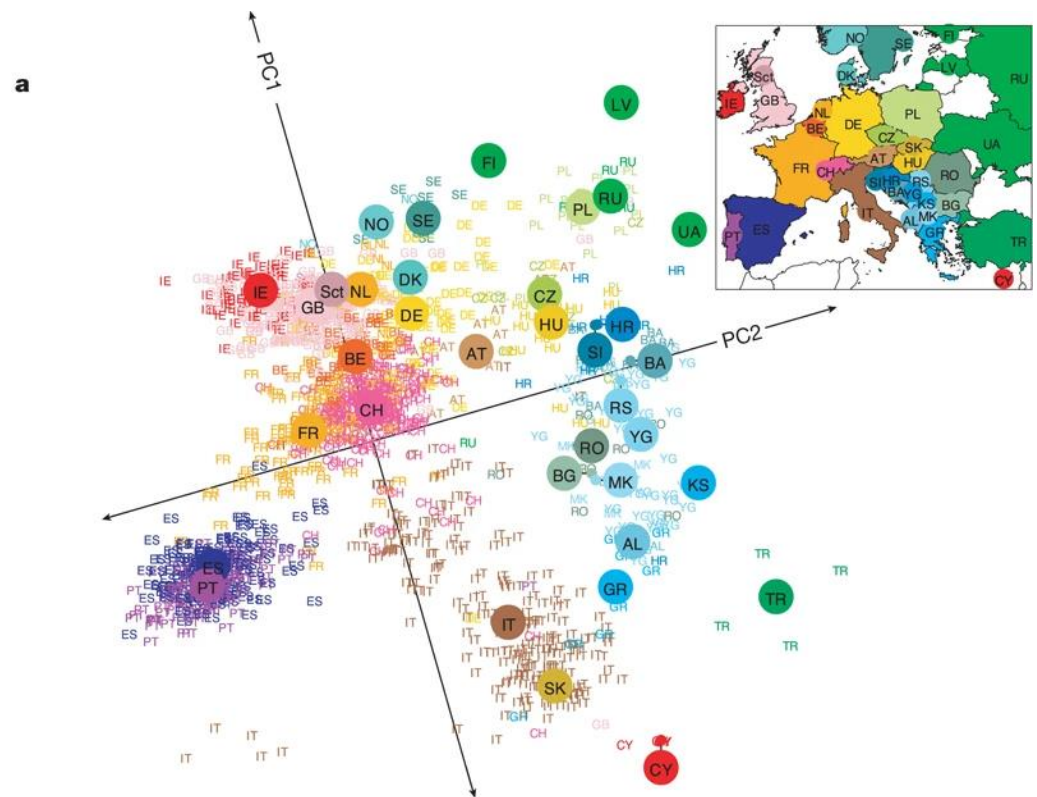
```
--recode vcf bgz
```

```
--output-chr
```

```
--out
```

Principal Component Analysis (PCA)

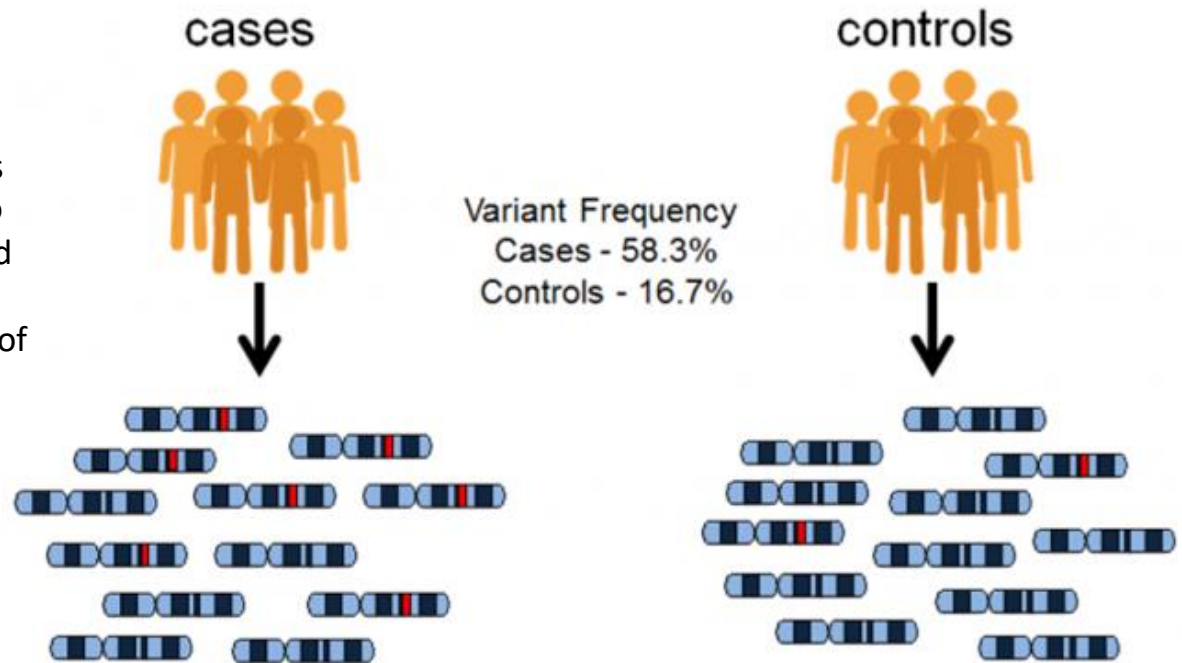
- PCA is a statistical method that reduces the complexity of datasets by finding the main patterns of variation.
- PCA is often used to examine population structure and ancestry patterns from genome-wide SNP data



10.1038/nature07331

Genome-Wide Association Studies (GWAS)

- Genome-Wide Association Studies (GWAS) scan the entire genome to identify genetic variants associated with diseases or traits
- Because we are making hundreds of thousands to millions of comparisons, we use a different p value threshold



<https://www.ebi.ac.uk/training/online/courses/gwas-catalogue-exploring-snp-trait-associations/what-is-gwas-catalog/what-are-genome-wide-association-studies-gwas/>

Manhattan Plots

